

Ross Jacobucci

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SUMMARY

Machine learning scientist with ten years leading research programs that turn smartphone, wearable, and text data into risk predictions and adaptive interventions for mental health and suicide prevention. Led \$7.6M in federal awards as PI or MPI and contributed to \$29M more as Co-Investigator. Built and ran an LLM fine-tuning and inference pipeline on a 15-GPU cluster; designed and analyzed randomized experiments including factorial and micro-randomized trials. Author of *Machine Learning for Social and Behavioral Research* (Guilford, 2023), 78 peer-reviewed papers, and R packages with 400K+ downloads.

SKILLS

Modeling and inference: predictive modeling and evaluation (regularized and tree-ensemble models, calibration, leakage and inflated-performance checks); experimentation and causal inference (factorial and micro-randomized trials, causal excursion effects, heterogeneous treatment effects with causal forests, policy learning, cross-cohort transport tests); reinforcement learning and bandits for adaptive interventions; longitudinal, multilevel, and time-series models; SEM and psychometrics; NLP and text mining.

LLMs: LoRA/PEFT fine-tuning of open models including vision-language models (Qwen); vLLM and SGLang inference; Unsloth; automated model-selection harnesses with grouped cross-validation, lockbox test sets, and promotion gates; OCR and vision-language classification at scale (7M+ images).

Languages and tools: Python (PyTorch, Hugging Face Transformers/PEFT, scikit-learn, pandas), R (CRAN package author), SQL, Mplus, Git, Linux/HPC; self-managed GPU servers.

Data: passive smartphone sensing, screenshots, wearables (Empatica, Garmin), ecological momentary assessment, clinical and survey data.

EXPERIENCE

Center for Healthy Minds, University of Wisconsin–Madison

Madison, WI

Research Associate Professor (2026–present); Research Assistant Professor (2024–2026)

2024–present

- Principal Investigator of a \$6.6M ARPA-H award (EVIDENT, 2026–2028), leading a team of 20 to measure rapid clinical change in behavioral health from multimodal data: designed the protocol and measurement stack for a 1,500-person randomized trial and a 300-person observational cohort combining Empatica and Garmin wearables, Android and iOS passive sensing, video, and EMA; selected vendors and data platforms; led IRB, legal, and data-security review.
- Built the lab's LLM stack: LoRA fine-tuning of open vision-language models (Qwen) with Hugging Face PEFT and Unsloth, vLLM/SGLang inference, and an automated model-selection harness (user-grouped cross-validation, lockbox test set, promotion gates) on a 15-GPU L40S cluster inside a restricted data environment.
- Processed 7.4M+ smartphone screenshots with OCR and vision-language classification on an HPC cluster to detect suicide-related language and predict momentary suicidal ideation (*JMIR Mental Health*, 2026).
- Led the causal and personalization analysis of a two-cohort factorial trial (2^4 and 2^6 designs, about 760 participants each) with a nested micro-randomized trial (380 participants, 10,000+ person-days): estimated causal excursion effects, tested heterogeneity with causal forests, policy trees, and cross-cohort transport, and delivered deployment rules for message timing and targeting; designed the Phase 2 bandit-driven JITAI and its simulation study.
- Showed that passively logged nighttime phone use predicts next-day suicide risk (*JAMA Network Open*, 2025) and linked screenshot-derived screen time to daily suicide risk (*npj Digital Medicine*, 2025).
- Co-Investigator on NIH R01, DARPA, and Templeton awards totaling \$29M; supervise a data scientist and research staff; responsible for participant-safety procedures in studies enrolling people at risk for suicide.

Department of Psychology, University of Notre Dame

Notre Dame, IN

Assistant Professor

2017–2024

- Founded a lab applying machine learning to suicide and self-injury research; MPI on two NIH R21 awards (\$860K) on real-time suicide risk detection from passively captured smartphone screenshots and on adaptive time sampling for momentary risk assessment.
- Developed prediction methods for text and self-report data and published a critique of inflated prediction performance in suicide machine learning (*Clinical Psychological Science*, 2021).
- Authored CRAN packages *regsem*, *MplusTrees*, and *longRPart2* (400K+ combined downloads) for regularized SEM and tree-based analysis of longitudinal data.

- Advised three PhD students in quantitative and clinical psychology; taught graduate regression and machine learning courses; co-authored *Machine Learning for Social and Behavioral Research*.

SELECTED PUBLICATIONS

78 peer-reviewed papers; full list at https://scholar.google.com/citations?user=K7_cclwAAAAJ.

- Jacobucci R, Shao W, Kobrinsky V, Ammerman B (2026). Predicting momentary suicidal ideation from smartphone screenshots using vision-language models. *JMIR Mental Health*.
- Jacobucci R, Jones SG, Blacutt M, Ammerman BA (2025). Passive vs active nighttime smartphone use as markers of next-day suicide risk. *JAMA Network Open*.
- Jacobucci R, Blacutt M, Ram N, Ammerman BA (2025). Smartphone screen time and suicide risk in daily life captured through high-resolution screenshot data. *npj Digital Medicine*.
- Ammerman BA, Kleiman EM, O'Brien C, . . . , Jacobucci R (2025). Smartphone-based text obtained via passive sensing as it relates to direct suicide risk assessment. *Psychological Medicine*.
- Jacobucci R, Littlefield A, Millner AJ, Kleiman EM, Steinley D (2021). Evidence of inflated prediction performance: A commentary on machine learning and suicide research. *Clinical Psychological Science*.
- Jacobucci R, Grimm KJ, McArdle JJ (2016). Regularized structural equation modeling. *Structural Equation Modeling*.

OPEN-SOURCE SOFTWARE

regsem (regularized SEM; 300K+ downloads), **MplusTrees**, **longRPart2**, **dtree**, **autoSEM** — R packages on CRAN, authored and maintained since 2016.

BOOK AND HONORS

- Jacobucci R, Grimm KJ, Zhang Z (2023). *Machine Learning for Social and Behavioral Research*. Guilford Press. Barbara Byrne Award for Outstanding Book, Society of Multivariate Experimental Psychology (2025).
- Early Career Research Award, Society of Multivariate Experimental Psychology (2023); Rising Star, Association for Psychological Science (2021); Anastasi Dissertation Award, APA Division 5 (2019).
- Research featured in *National Geographic* (2025), *APA Monitor* (2023), and *APS Observer* (2022).

TEACHING AND PROFESSIONAL ACTIVITIES

- Instructor, Applied Deep Learning with Python (2024) and Advanced Machine Learning (2021–2024), Statistical Horizons; instructor, APA Advanced Training Institutes (2016–2020).
- Editorial boards: *Psychological Methods*, *Clinical Psychological Science*, *Psychological Science*. NIMH grant review panels (2020–2023).
- 15 invited talks on AI and suicide prevention, screenshot-based prediction, and regularized SEM, including USC, Stanford, and the Max Planck Institute for Human Development.

EDUCATION

Ph.D., Psychology (Quantitative Methods), University of Southern California, 2017. NIH T32 predoctoral fellow.
B.A., Psychology, Luther College, 2010.